Seyyed Ali Aghamiri

LinkedIn | Email | Website | GitHub | Google Scholar

School of Electrical and Computer Engineering, University of Tehran · Tehran, Iran
Email: aghamiri.seyyedali@gmail.com

Research Interests

Distributed, Networked, and ML Systems; Resource Scheduling & Optimization

Education

M.Sc. in Computer Engineering - Computer Architecture Sep 2023 - Present
University of Tehran
GPA: 17.15/20.00
Supervisor: Prof. Ahmad Khonsari
Thesis: Optimization of tip selection algorithm on DAG based blockchain in IOT.
Relevant coursework: Advanced Computer Networking, Performance Evaluation of Computer Systems, Cryptocurrencies, Internet Measurement, Cyber-Physical Systems (CPS).

B.Sc. in Computer Engineering - Software Engineering Sep 2017 - Aug 2022
Persian Gulf University
GPA: 15.61/20.00

Publications

[J1] **Seyyed Ali Aghamiri**, A. Karimi, A. Khonsari, and M. Dolati, "Compute and IO-Aware Distributed Training Management in Converged Networks," IEEE Transactions on Cloud Computing, submitted.

[C2] M. Mozhganfar, P. Jamshidi, **Seyyed Ali Aghamiri**, M. Ghasemi, M. Dolati, F. Tashtarian, A. Khonsari, and C. Timmerer, "YTLive: A Dataset of Real-World YouTube Live Streaming Sessions," IEEE/IFIP Network Operations and Management Symposium (NOMS), 2026.

[C1] **Seyyed Ali Aghamiri**, R. Sharifnia, and A. Khonsari, "Utilizing Transaction Prioritization to Enhance Confirmation Speed in the IOTA Network," 29th International Computer Conference, Computer Society of Iran (CSICC), 2025.

Research Experience

Distributed Training and ML Systems 2025 - Present
University of Tehran; advised by Dr. Mahdi Dolati
Developed and implemented the optimization and heuristic components of NEXUSTRAIN, a resource-aware multi-job distributed training scheduler that jointly considers hybrid data/pipeline parallelism, GPU placement, compute scheduling, remote-data prefetching, and shared network/storage constraints.

Formulated the scheduling problem as a time-indexed mixed-integer linear program and developed scalable search and resource-aware scheduling procedures.

Implemented the solver and heuristic framework, adapted the TopoOpt/FlexNetPacket-based simulation environment, and conducted numerical and packet-level evaluations.

IOTA / DAG-Based Distributed Ledgers 2023 - Present
University of Tehran; Supervisor: Prof. Ahmad Khonsari
Investigating tip-selection optimization in IOTA/DAG-based distributed ledgers, with a current focus on mitigating parasite-chain attacks.

Developed and evaluated a priority-based tip-selection mechanism using the official IOTA private Tangle and Hornet node software to reduce confirmation time for high-priority transactions.

L4S for Low-Latency Video Streaming 2023 - 2024

Research Project with ATHENA / University of Klagenfurt

Investigated the use of Low Latency, Low Loss, Scalable Throughput (L4S) for low-latency video-streaming scenarios.

Set up and experimented with a Mininet-based environment to study L4S behavior and low-latency transport mechanisms.

QoE Modeling in Volumetric Video Streaming 2025 - 2026

Research Project with Klagenfurt/ATHENA and IMDEA Networks

Contributing to an ongoing survey on end-to-end Quality of Experience (QoE) modeling for volumetric video streaming.

Responsible for reviewing and synthesizing subjective QoE assessment studies, including assessment methodologies, interaction settings, streaming impairments, and dataset availability.

Teaching & Academic Leadership

Chief Teaching Assistant / Teaching Assistant 2024 - 2025

University of Tehran

Chief TA - Distributed Systems: led a 6-member TA team for approximately 90 undergraduate students; coordinated project and exam design, grading, and course logistics.

Chief TA – Blockchain Theory: contributed to exam design, grading, student support and technical troubleshooting.

TA - Network Laboratory (two semesters): independently ran one weekly lab section, evaluated experiments, and supported students during hands-on networking exercises.

TA - Internet Measurement: assisted with course assignments and student support.

TA - Design Validation for HDL Models: assisted with project design, project evaluation, and student support.

Research Mentoring

Undergraduate Research Mentor 2024 - Present

University of Tehran

Mentored three undergraduate research projects, from scoping and technical onboarding through implementation and evaluation.

Projects included collecting data from the IOTA network, RL-based tip selection for parasite-chain detection, and distributed training systems research.

Guided one student from initial distributed-learning concepts to implementation and manuscript contribution as the second author on the submitted TCC manuscript.

Technical & Laboratory Infrastructure

Network Laboratory Infrastructure Project

University of Tehran

Installed VMware ESXi on a new laboratory server and integrated hosts through vSphere for centralized virtual-machine management.

Deployed Prometheus-based server monitoring and configured Grafana dashboards for resource and system metrics.

Deployed and administered PNETLab, created course sections and environments, and trained students in its use.

Migrated and adapted laboratory exercises from GNS3 to PNETLab and consolidated the revised experiments into a unified lab manual.

Entrepreneurship & Outreach

Co-Founder & Product Manager

Cryptocurrency Exchange Startup

Co-founded a cryptocurrency exchange startup and led product planning, competitor analysis, feature comparison, and product prioritization.

Delivered public workshops for roughly 50-100 participants and university seminars covering blockchain fundamentals, on-chain concepts, and digital-asset investment basics.

Selected Technical Projects

Blockchain Node & Testnet Deployment
Graduate course project - University of Tehran
Implemented a blockchain node, deployed it on a test network, and generated a test transaction while studying Bitcoin-style transaction and node workflows.
Code: github.com/ali-aghamiri/cryptocurrency-coursework

Internet Measurement & Monitoring
Graduate course project - University of Tehran
Deployed Elasticsearch for system monitoring and configured Grafana components for metric visualization and operational inspection.
Code: github.com/ali-aghamiri/internet-measurement-coursework

Independent Distributed Machine Learning Study
Ongoing self-study
Building four independent Distributed ML study labs (MPI/FedAvg, CUDA/DDP, Spark, and torchrun/Accelerate); local CPU paths validated, with GPU and cluster experiments ongoing.
Code: github.com/ali-aghamiri/distributed-ml-study-labs

Technical Skills

Systems & Infrastructure: Linux, Docker, Docker Compose, VMware ESXi, vSphere, PNETLab, Mininet, Git/GitHub

Programming: Python, C; Go (coursework); ML systems (current study): NumPy, PyTorch, MPI/FedAvg, DDP, PySpark, CUDA, Accelerate

Monitoring / Experimentation: Elasticsearch, Grafana, IOTA/Hornet; experience adapting and extending research simulators

References

Prof. Ahmad Khonsari
Professor, School of Electrical and Computer Engineering
University of Tehran, Iran
a_khonsari@ut.ac.ir

Dr. Mahdi Dolati
Assistant Professor, Department of Computer Engineering
Sharif University of Technology, Iran
mdolati@sharif.edu